

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: CIV101

Course: **BASIC CIVIL ENGINEERING**

QP No. :U004R-1

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 marks

1. When a building reached the end of its design life, it was decided to demolish it. In this regard, how will the building code protect neighboring structures?
2. Name any two tests that can be recommended to ensure the quality of the cement.
3. Forest degradation is quite common in all over India. In this scenario what is your recommendation for the alternate materials in order to preserve trees? *Bamboo, wood products, Recycled materials, Alternative construction techniques, Timber plantation*
4. Differentiate a cut lintel from a continuous lintel. In what circumstances would you prefer these two structural components for a building?
5. Suggest a staircase (a) for a residential building with a very limited set back distance (b) for a commercial building with a high set back distance.

6/ Make a recommendation to the highway department in your town to address the issue of vehicle overturning, particularly in curves. Justify your answer with an appropriate diagram.

7/ Name four surveying tools that could be used in a feasibility study for a large commercial project.

8/ Enlist the different kinds of drawings (maps) that are utilised in the administrative office of the district

9/ Why does the Irrigation sector require more attention from the government in a developing country like India?

10/ The percentage of total water that is made up by groundwater is 0.68 percent. How did the pollution of ground water become such a common practice in our country?

PART - B

Answer all the questions

4 x 15 = 60 marks

11/ It was decided to build a multi-speciality hospital on the outskirts of Chennai, and how people from various disciplines within civil engineering should collaborate to complete the project.

(OR)

12/ RCC is a combination of various building materials -- What is the significance of each material used in it? Describe the various types of concrete used in the construction sector.

13/ For a load bearing structures, Stones are used for foundations, while bricks are used for superstructures. Justify this statement with relevant examples of stone masonry and brick masonry.

(OR)

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14/ (a) How many steps can be accommodated in each flight of a dog legged staircase if the building's overall height is 3.20 m? Assume the step's rise and tread are 0.15 m and 0.30 m, respectively. Draw a suitable illustration to accompany your response. (8)

(b) With the aid of suitable drawing, Explain the various components of a National Highway in India. (7)

15/ The following consecutive reading were taken in with a levelling instrument on ground. The Reduced Level of first point is 100.00. Find the level difference using HCL method. The readings are 1.350, 1.150, 1.855, 0.250, 1.500, 1.750, 2.185, 2.205, 1.350, 0.950 and 2.350. The instrument was shifted after 5th and 9th reading.

(OR)

16/ (a) With suitable drawing, explain the two main principles associated with surveying (5)

(b) What is local attraction? How is it detected and removed? (5)

(c) Write a note on Trapezoidal rule and Simpson's rule for the calculation of areas. (5)

17/ How would using rainwater conservation techniques increase the water's ability to survive? How will you personally implement such a practice in your lifetime? Drawings that are appropriate should be used to support the conservation activities.

(OR)

18/ (a) Draw a dam's cross section and describe the forces at work on it. (5)

(b) List the advantages and disadvantages of gravity dam (5)

(c) Point forth the benefits of infrastructure like dams for humanity. (5)

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PART - C

Answer the following

1 x 20 = 20 marks

19. If you plan to build your own home in 2030, create a suitable framework to incorporate the planning principles into your design. For the same building, how your understanding on various building material properties, water conservation and surveying will assist you in completing the construction with uncompromised quality and durability for environmentally sustainable project.
